



SMART-Edit: Spectral Mutual Attention Refinement and fusion for precise training-free image editing

Jili Chen^a, Yaxin Tu^b, Qionghao Huang^a, Xizhe Wang^a, Xiaodi Huang^c,
Changqin Huang^{b,a}*

^a Zhejiang Key Laboratory of Intelligent Education Technology and Application, Zhejiang Normal University, Jinhua, 321004, China

^b College of Education, Zhejiang University, Hangzhou, 310058, China

^c School of Computing, Mathematics and Engineering, Charles Sturt University, Albury, NSW, 2640, Australia

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ABSTRACT

Image generation and editing based on Multimodal Diffusion Transformers (MM-DiT) have emerged as the mainstream paradigm. However, for training-free image editing, directly applying existing U-Net-based semantic control methods to MM-DiT inevitably causes severe semantic leakage and imprecise editing localization, resulting in unintended modifications. Therefore, this paper proposes Spectral Mutual Attention Refinement and Transition (SMART-Edit), a novel training-free image editing framework explicitly tailored for MM-DiT. We begin by revisiting the joint attention mechanism in MM-DiT. To mitigate semantic entanglement and initially localize the editing target, we introduce the Dual Attention Refinement (DAR) module, which performs strict semantic decoupling using textual and visual self-attention on both the text-to-image and image-to-text cross-attentions. Subsequently, these refined cross-attentions are fused in the frequency domain via the Spectral Mutual Attention Fusion (SMAF) module to precisely capture the spatial layout of the target object. To transform this localized attention into an editing-friendly mask, we further propose the Soft Mask Transition (SMT) module to ensure natural and continuous boundary edges. SMART-Edit seamlessly integrates into existing training-free editing pipelines with negligible additional time overhead. Extensive experiments on Flux.1 Dev and SD3/3.5 demonstrate that our framework elevates editing faithfulness, background preservation, and overall aesthetics. Furthermore, attention analysis confirms that it significantly improves precise semantic localization during both the inversion and denoising processes.

1. Introduction

In recent years, the rapid advancement of large-scale text-to-image generative models [1,2] has driven text-guided image editing to become a pivotal application. The core objective of text-guided image editing is to precisely modify specific target attributes according to the user prompt while strictly preserving the structural layout and irrelevant background. To achieve this, training-based methods typically fine-tune the generative models [3–5] to enable instruction-following image editing. However, these approaches necessitate extensive curated datasets and substantial computational resources, resulting in time-consuming and resource-intensive training pipelines.

To circumvent these limitations, many training-free image editing methods have emerged as compelling alternatives. On the one hand, some inversion-based techniques invert the source image into a semantically aligned latent representation conditioned on a source

prompt, and subsequently steer the editing process by denoising the latent representation with a target prompt, typically relying on feature injection [6–8] or ODE trajectory calibration [9,10] to preserve structural consistency. On the other hand, inversion-free approaches [11,12] bypass the inversion step, instead steering the source latent toward the target distribution through velocity differences in the vector field. Besides, extensive studies [13] have focused on achieving fine-grained spatial decoupling. These methods typically utilize image-to-text cross-attention maps within U-Nets or Multimodal Diffusion Transformers (MM-DiT) to derive spatial masks, which restrict editing to target regions while preserving non-target regions.

Despite these advancements, existing attention-based semantic localization for image editing methods suffer from severe semantic leakage when applied to MM-DiT, as illustrated in Fig. 1. Consequently, the generated masks (e.g., “flowers” in Fig. 1) often exhibit unclear

* Corresponding author at: College of Education, Zhejiang University, Hangzhou, 310058, China.

E-mail addresses: irelia@zjnu.edu.cn (J. Chen), tuyx1996@zju.edu.cn (Y. Tu), qhhuang@m.scnu.edu.cn (Q. Huang), xzwang@zjnu.edu.cn (X. Wang), xhuang@csu.edu.au (X. Huang), cqhuang@zju.edu.cn (C. Huang).

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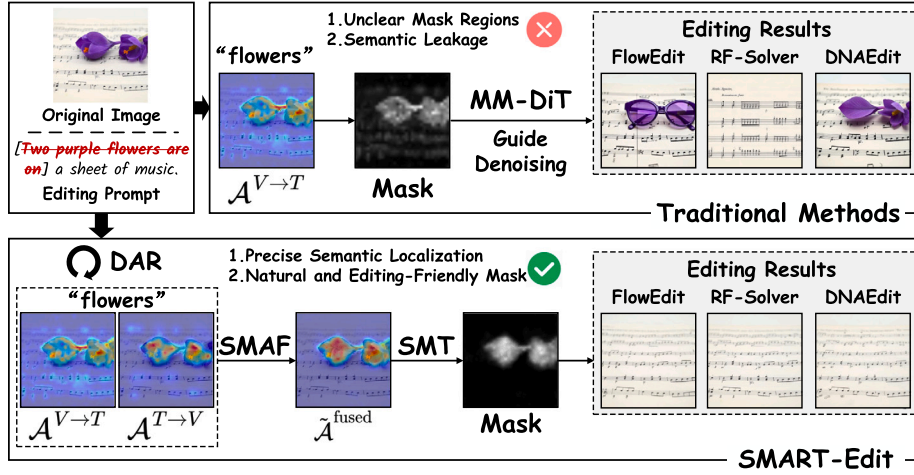


Fig. 1. SMART-Edit generates more precise and suitable masks to guide training-free image editing compared to traditional methods. Traditional methods [14–16] typically rely solely on extracting image-to-text ($A^{V \rightarrow T}$) attention in MM-DiT for editing control. While this is a common practice in U-Net architectures, it is not fully adaptable to MM-DiT, often leading to unclear mask regions and semantic leakage. Consequently, when integrated with other editing methods (e.g., FlowEdit [11], RF-Solver [10], and DNAEdit [12]), they often fail to effectively realize the desired editing intentions. We discover that text-to-image attention ($A^{T \rightarrow V}$) can also be effectively utilized to provide semantic localization guidance. By employing a refinement and spectral fusion mechanism for dual attention ($A^{V \rightarrow T}$, $A^{T \rightarrow V}$) and further diffusing it into natural transitions, SMART-Edit generates highly precise mask regions. Furthermore, SMART-Edit is an architecture- and method-agnostic, plug-and-play module that achieves precise semantic alignment.

boundaries, leading to unintended artifacts, background corruption, and incomplete edits. Fundamentally, this limitation stems from the inherent joint attention mechanism of MM-DiTs. Concatenating text and image tokens into a unified sequence for holistic interaction inevitably causes cross-modal feature coupling, exacerbates token competition, and disrupts the intrinsic attention distribution. Crucially, the joint attention mechanism in MM-DiTs compounds the semantic leakage typical of U-Net-based methods [17], rendering them highly unreliable for precise spatial control. Motivated by these observations, a natural question arises: **(1) The Semantic Entanglement Problem: How can the semantically entangled joint attention mechanism in MM-DiTs be effectively disentangled to achieve precise semantic localization and prevent spatial leakage in training-free image editing?** However, even with precise localization, hard thresholding of attention maps for editing remains problematic. This discretization forces rigid boundaries, inducing artifacts and restricting editability due to abrupt transitions between the edited region and the background. This naturally leads to our second critical question: **(2) The Boundary Discretization Problem: How can localized semantic attentions be transformed into smooth, structurally coherent, and editing-friendly spatial masks that enable precise image editing?**

To address the aforementioned challenges, we present Spectral Mutual Attention Refinement and Transition (SMART-Edit), a method-agnostic framework tailored for training-free image editing in MM-DiTs (Fig. 1). To tackle Q1, we revisit the image-to-text ($A^{V \rightarrow T}$) and text-to-image ($A^{T \rightarrow V}$) cross-attentions within the MM-DiT blocks. We observe that while both individual attention branches inevitably suffer from varying degrees of semantic leakage, their intrinsic cross-modal interaction mechanisms encapsulate highly complementary spatial cues. Motivated by this insight, we propose a **Dual Attention Refinement (DAR)** module to independently optimize both cross-attention maps using textual and visual self-attention maps within the MM-DiT, effectively mitigating their distinct leakage patterns. Subsequently, we introduce a **Spectral Mutual Attention Fusion (SMAF)** module to seamlessly integrate these two refined attention maps in the frequency domain to yield highly precise localized regions. Furthermore, to answer Q2, SMART-Edit incorporates a **Soft Mask Transition (SMT)** module to transform abrupt spatial boundaries into continuous, natural transitions. SMT begins with an intensity-weighted semantic centroid anchor. Coupled with morphological refinement, it iteratively expands the mask

based on attention homogeneity. The SMT module effectively eliminates unnatural hard artifacts while preserving a seamless blending mask. As a method-agnostic framework, SMART-Edit can be integrated into various state-of-the-art training-free editing frameworks, boosting the capacity of semantic localization and overall editing quality. Our main contributions can be summarized as follows:

- **Method-Agnostic and Training-Free Image Editing Framework.** We propose SMART-Edit, a plug-and-play structural guidance framework for image editing. It seamlessly integrates with diverse MM-DiT backbones (e.g., Flux.1 Dev, SD3/3.5) and training-free image editing methods (e.g., inversion-based RF-Solver and inversion-free FlowEdit) to provide highly accurate spatial control without requiring any parameter optimization.
- **Precise Semantic Localization and Editing for MM-DiTs.** We propose a novel “localize-then-edit” paradigm for MM-DiTs. By integrating DAR to mitigate semantic leakage, SMAF to extract structurally aligned semantic cues, and SMT to ensure natural boundaries, we robustly convert raw attention maps into precise spatial masks for accurate semantic localization and editing with negligible additional time overhead.
- **Overall Enhancement of Editing Quality.** Extensive experiments on PIE-Bench and EditEval demonstrate that SMART-Edit enhances existing training-free image editing baselines. SMART-Edit yields significant improvements across background preservation, editing faithfulness, and overall visual aesthetics.

2. Related work

2.1. Image editing with diffusion models

Image editing aims to modify specific image attributes while maintaining structural and content consistency. Leveraging powerful generative priors, diffusion models [18,19] have emerged as the leading framework for advanced image editing. To balance editing precision and structure preservation, research has branched into training-based and training-free paradigms. Training-based methods involve end-to-end fine-tuning of the entire network [3], specific layers [20], or embeddings [21–24] to maintain the structural integrity and content consistency of the reference image. In contrast, training-free paradigms have gained significant attention for their computational efficiency

and resource-friendly attributes. These methods utilize inversion techniques (e.g., DDIM [25]) to map images into a structurally informative noise space for structure preservation [26,27]. Building upon diffusion models, recent research has shifted toward integrating Rectified Flow [28] with Diffusion Transformer (DiT) [2] architectures, aiming to enhance editing scalability and efficiency. Specifically, some approaches [12,29,30] focus on mitigating inversion errors to better encapsulate structural semantics; others achieve fine-grained control by directly manipulating the attention mechanisms [31]. Furthermore, inversion-free approaches [11] guide the editing process by computing the velocity difference between target and source vector fields, without requiring inversion. Despite their advancement, existing methods often lack spatial guidance, leading to unintended modifications in non-target regions. To address this limitation, we empower the model to automatically localize target regions, strictly confining modifications to the intended areas without affecting the background.

2.2. Semantic control for visual editing

Early works in text-to-image generation achieved semantic control by manipulating and replacing activation maps within cross-attention [32] and self-attention [13] layers of the U-Net architecture. In flow matching, this paradigm has been further adapted, where specific QKV components from the inversion trajectory are injected into the denoising process to maintain structural and semantic consistency [7,8,33]. Previous studies have established the critical importance of semantic localization for both image generation and editing [34]. Therefore, considerable research has emphasized attention-based mask generation, allowing for region-specific editing that effectively maintains content consistency across the unedited background [35]. However, when applied to modern MM-DiT backbones, existing techniques expose two fundamental limitations. First, they often directly transfer semantic localization strategies designed for U-Net architectures. This fails to account for the unified attention mechanism in MM-DiTs, which inevitably causes “semantic leakage” and makes it inherently challenging to disentangle specific regions for modification. Second, directly converting raw attention maps into masks via hard thresholding often introduces unnatural edge artifacts, compromising the overall visual aesthetics. To address these limitations, we embrace adaptive structural modeling and multi-source feature fusion [36–38] to develop a semantic localization framework tailored for MM-DiTs. By refining frequency-domain mutual attention maps into soft masks, our method achieves precise semantic localization and high-fidelity image editing.

3. Preliminaries

3.1. Training-free image editing

Rectified Flow [28] establishes a deterministic mapping between data $x_0 \sim \pi_0$ and noise $x_1 \sim \pi_1$ via the ODE $\frac{dx_t}{dt} = v_\theta(x_t, t)$, where the velocity field v_θ is optimized to follow the straight path $x_t = (1-t)x_0 + tx_1$. Based on this deterministic formulation, current training-free image editing paradigms can be categorized into two distinct branches: **inversion-based** and **inversion-free** approaches.

Inversion-based Editing. This framework enables high-fidelity inversion by integrating the ODE forward from $t = 0$ to 1 conditioned on the source prompt p_{src} , transforming the real image x_0 into an inverted latent representation $\tilde{x}_1$:

$$\tilde{x}_1 = x_0 + \int_0^1 v_\theta(x_t, t, p_{\text{src}}) dt, \quad (1)$$

subsequently, inversion-based editing is achieved by solving the reverse-time ODE, i.e., the denoising process, from $t = 1$ to 0, conditioned on the inverted latent and the target prompt p_{tar} :

$$\tilde{x}_0 = \tilde{x}_1 + \int_1^0 v_\theta(x_t, t, p_{\text{tar}}) dt, \quad (2)$$

where $\tilde{x}_0$ represents the final edited image. By strategically injecting intermediate features from the inversion process into the denoising phase, this framework can further facilitate precise semantic manipulation while strictly preserving the original structure.

Inversion-free Editing. In contrast, inversion-free editing directly steers the generative trajectory starting from a sampled noise state $x_1 \sim \pi_1$. To align these dynamics for precise semantic manipulation, this approach modifies the source generative flow by injecting an instantaneous velocity residual:

$$\Delta v_\theta(x_t, t) = v_\theta(x_t, t, p_{\text{tar}}) - v_\theta(x_t, t, p_{\text{src}}), \quad (3)$$

the final edited image $\tilde{x}_0$ is then obtained by integrating this manipulated velocity field backward from $t = 1$ to 0. This method enables efficient editing while completely circumventing the computational overhead of explicit ODE inversion.

3.2. Attention mechanism in MM-DiT

Modern architectures such as Flux.1 Dev transition from UNet-based designs to MM-DiT [39]. MM-DiT utilizes a joint self-attention mechanism to integrate textual and visual information within each layer. In an MM-DiT block, textual embeddings T and visual embeddings V are mapped into a shared latent space. To maintain modality-specific features, the model employs independent learnable projection matrices: $Q_T = TW_Q^T, K_T = TW_K^T, V_T = TW_V^T$ for the textual branch and $Q_V = VW_Q^T, K_V = VW_K^T, V_V = VW_V^T$ for the visual branch. The core interaction is driven by concatenating queries and keys from both modalities (denoted by $\oplus$):

$$\mathcal{A}_{\text{joint}} = \text{Softmax} \left(\frac{[Q_T \oplus Q_V][K_T \oplus K_V]^T}{\sqrt{d}} \right), \quad (4)$$

where the joint attention matrix $\mathcal{A}_{\text{joint}} \in \mathbb{R}^{H \times (N+M) \times (N+M)}$ models the complete bidirectional interactions between the two modalities. Here, N and M denote the sequence lengths of the textual and visual tokens, respectively, and H denotes the number of attention heads in MM-DiT. To obtain a unified spatial representation, we average $\mathcal{A}_{\text{joint}}$ across all H heads. The resulting 2D matrix is then partitioned into four distinct quadrants: (1) $\mathcal{A}^{T \rightarrow T} \in \mathbb{R}^{N \times N}$ captures internal dependencies and competition between text tokens. (2) $\mathcal{A}^{V \rightarrow V} \in \mathbb{R}^{M \times M}$ encodes the spatial affinity and structural layout of the image. (3) $\mathcal{A}^{V \rightarrow T} \in \mathbb{R}^{M \times N}$ maps visual regions to their corresponding semantic tokens, providing region-conditioned semantic attribution and precise localization cues. (4) $\mathcal{A}^{T \rightarrow V} \in \mathbb{R}^{N \times M}$ associates textual concepts with visual regions involved in feature generation, providing concept-conditioned localization signals. Implementation details of attention extraction are provided in Appendix B.

4. Methodology

In this section, we introduce **Spectral Mutual Attention Refinement and Transition (SMART-Edit)**, a training-free image editing framework inspired by the intrinsic attention mechanism of MM-DiT. The core of our method lies in the semantic localization guided by the SMART module. We begin by extracting and refining the intrinsic multimodal attention maps of MM-DiT via Dual Attention Refinement (Section 4.1). Subsequently, we introduce an attention integration strategy using Spectral Mutual Attention Fusion (Section 4.2), followed by a Soft Mask Transition process to yield a spatially precise and natural mask (Section 4.3). Finally, we detail the plug-and-play mechanism of SMART when integrated into other image editing pipelines (Section 4.4). The overall process of SMART-Edit and the detailed pipeline of the SMART module are illustrated in Fig. 2 and Fig. 3, respectively.

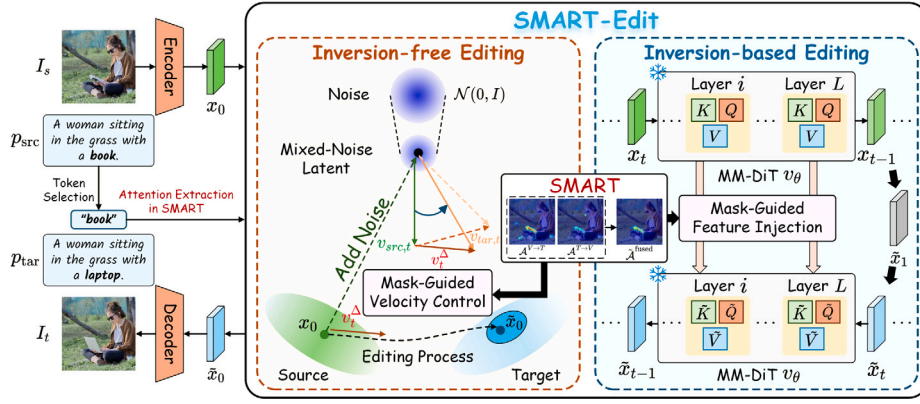


Fig. 2. Overview of the SMART-Edit framework. The core SMART module can be integrated with existing training-free methods in a plug-and-play manner to achieve highly precise image editing.

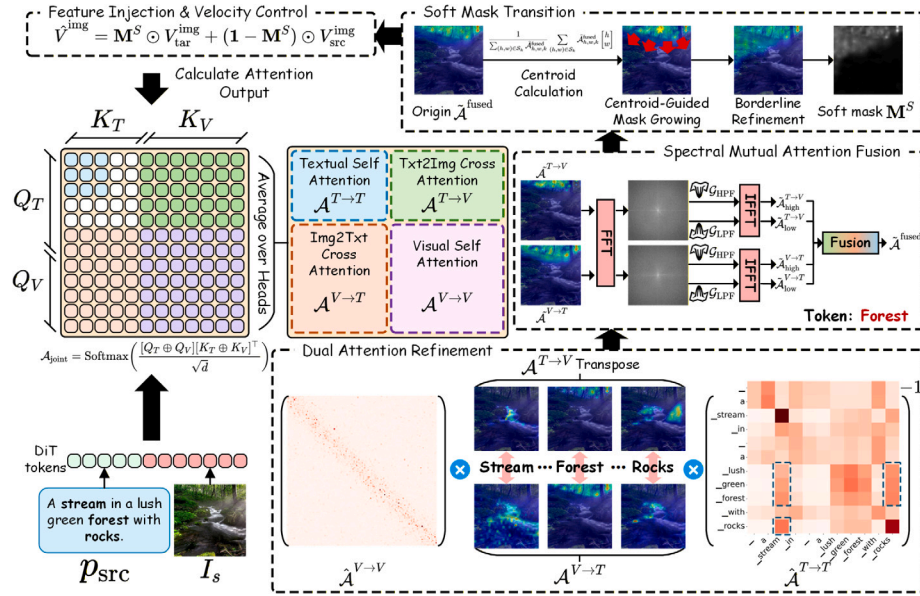


Fig. 3. Detailed pipeline of the SMART module. To achieve precise semantic guidance, SMART first refines the $\mathcal{A}^{T \rightarrow V}$ and $\mathcal{A}^{V \rightarrow T}$, respectively. Notably, the blue dashed boxes in $\mathcal{A}^{T \rightarrow T}$ highlight the potential attention competition among tokens. These refined maps are then fused in the frequency domain and processed via a centroid-guided soft mask transition. Notably, the resulting mask can be directly integrated into other training-free methods via feature injection or velocity control.

4.1. Dual attention refinement

Conventional methods typically rely exclusively on $\mathcal{A}^{V \rightarrow T}$ for semantic localization. However, our key insight is that both cross-modal attentions in MM-DiT ($\mathcal{A}^{V \rightarrow T}$ and $\mathcal{A}^{T \rightarrow V}$) encapsulate complementary localization cues.

Insights: Specifically, $\mathcal{A}^{V \rightarrow T}$ captures *region-to-concept* attribution. Theoretically, it functions as a bottom-up semantic query, modeling the probability of each visual patch affiliating with specific textual concepts. While it captures fine-grained localized details, it is highly susceptible to semantic leakage, where a visual patch may loosely attend to contextually related but unintended words. In contrast, $\mathcal{A}^{T \rightarrow V}$ emphasizes *concept-to-region* localization. It acts as a top-down spatial broadcast, dictating how a text token distributes its semantic priors across the entire image grid. Although it establishes a robust global layout, it often yields coarse and overly expansive activation maps due to visual feature entanglement. Consequently, a unilateral reliance on either mechanism leads to a suboptimal spatial representation. A reliable semantic correspondence should remain consistent under this bidirectional transition. By enforcing bidirectional consistency, semantic correspondences that are consistently reflected in both attention

pathways are amplified, whereas mismatched or weakly supported associations are attenuated.

Although dual attentions provide complementary localization cues, prior work [33,40] has pointed out that the joint attention in MM-DiT leads to: (1) incomplete activation regions containing indistinct boundaries and (2) semantically irrelevant spurious activations, which introduce unrelated objects and extraneous artifacts.

To address these issues, we leverage the inverse textual self-attention matrix to mitigate spurious inter-textual correlations that trigger improper activations (as highlighted by the blue dashed boxes in Fig. 3). The underlying intuition is that the textual self-attention matrix serves as a proxy for the semantic dependencies induced by pretraining. For example, as illustrated in Fig. 3, *forest* and *rocks* frequently co-occur in descriptions such as “a lush green forest with rocks”. Under maximum-likelihood training, such recurring co-occurrence patterns encourage these concepts to become strongly coupled and biased in the learned representations, which constitutes a major source of semantic leakage when establishing text-visual correspondences. Consequently, when the model is instructed to localize the visual attributes associated with *rocks*, the resulting $\mathcal{A}^{T \rightarrow V}$ and $\mathcal{A}^{V \rightarrow T}$ may propagate the

localization signal to the correlated concept *forest* or even to green objects associated with it, as observed in [40], resulting in unintended activations. To formalize this intuition, let $C^{V \rightarrow T}$ denote the ideal direct correspondence between visual regions and textual concepts, without the semantic mixing introduced by inter-textual dependencies. The observed cross-modal attention $\mathcal{A}^{V \rightarrow T}$ can then be viewed as a composition of this direct correspondence with the textual semantic mixing, i.e., $\mathcal{A}^{V \rightarrow T} \approx C^{V \rightarrow T} \mathcal{A}^{T \rightarrow T}$. This perspective motivates multiplying by $(\mathcal{A}^{T \rightarrow T})^{-1}$ to approximately recover the direct correspondence $C^{V \rightarrow T}$ and thereby suppress semantic leakage. In this sense, the inverse textual self-attention matrix serves to counteract the semantic mixing induced by these pretraining dependencies and recover a less entangled correspondence. Concurrently, we utilize visual self-attention to sharpen object boundaries.

Specifically, we first reweight the visual and textual attention matrices, $\mathcal{A}^{V \rightarrow V}$ and $\mathcal{A}^{T \rightarrow T}$, transforming them into row-stochastic matrices $\hat{\mathcal{A}}^{V \rightarrow V}$ and $\hat{\mathcal{A}}^{T \rightarrow T}$. In practice, directly inverting $\hat{\mathcal{A}}^{T \rightarrow T}$ may be numerically unstable when the matrix is ill-conditioned. We therefore add a small diagonal regularization term λI before inversion, with $\lambda = 10^{-6}$, which improves numerical stability while introducing negligible perturbation to the original attention structure. Then, a comparison between p_{src} and p_{tar} is conducted to pinpoint the elements to be altered, yielding the sequence of modified elements $\mathcal{I}$. By indexing the attention matrices along the textual dimension, we derive the dual refined attention $\tilde{\mathcal{A}}^{V \rightarrow T}$ and $\tilde{\mathcal{A}}^{T \rightarrow V}$ as follows:

$$\tilde{\mathcal{A}}^{V \rightarrow T} = \text{norm}(\hat{\mathcal{A}}^{V \rightarrow V} \cdot \text{Select}_{\mathcal{I}}[\mathcal{A}^{V \rightarrow T} \cdot (\hat{\mathcal{A}}^{T \rightarrow T})^{-1}]), \quad (5)$$

$$\tilde{\mathcal{A}}^{T \rightarrow V} = \text{norm}(\hat{\mathcal{A}}^{V \rightarrow V} \cdot \text{Select}_{\mathcal{I}}[(\mathcal{A}^{T \rightarrow V})^{\top} \cdot (\hat{\mathcal{A}}^{T \rightarrow T})^{-1}]). \quad (6)$$

where $(\mathcal{A}^{T \rightarrow V})^{\top} \in \mathbb{R}^{M \times N}$ represents the transposed text-to-image attention matrix. The operator $\text{Select}_{\mathcal{I}}[\cdot]$ extracts the columns matching the text tokens designated for modification from both $\mathcal{A}^{V \rightarrow T}$ and $(\mathcal{A}^{T \rightarrow V})^{\top}$, yielding the dual refined attention maps $\tilde{\mathcal{A}}^{V \rightarrow T} \in \mathbb{R}^{M \times |\mathcal{I}|}$ and $\tilde{\mathcal{A}}^{T \rightarrow V} \in \mathbb{R}^{M \times |\mathcal{I}|}$, respectively. Fundamentally, multiplying by $(\hat{\mathcal{A}}^{T \rightarrow T})^{-1}$ mathematically offsets the semantic entanglement responsible for spurious activations, while leveraging $\hat{\mathcal{A}}^{V \rightarrow V}$ regulates spatial affinity to eliminate internal holes and blurry boundaries simultaneously.

4.2. Spectral mutual attention fusion

Despite both $\tilde{\mathcal{A}}^{V \rightarrow T}$ and $\tilde{\mathcal{A}}^{T \rightarrow V}$ encoding cross-modal alignments, they fundamentally operate on asymmetrical semantic levels [41], which leads to substantial divergence in their frequency bands and numerical distributions. Thus, a straightforward weighted sum fails to effectively fuse the two attention maps for semantic localization. Inspired by frequency-domain learning [42], we propose Spectral Mutual Attention Fusion (SMAF) to dynamically reconcile these spectral and scale mismatches.

To formalize the frequency-domain alignment, we first reshape the refined attention maps into $\tilde{\mathcal{A}}^{V \rightarrow T}, \tilde{\mathcal{A}}^{T \rightarrow V} \in \mathbb{R}^{H \times W \times |\mathcal{I}|}$, where $H \times W = M$, strictly maintaining the spatial aspect ratio of the input image. The 2D Fast Fourier Transform (FFT) is then performed along the first two spatial dimensions of these two cross-modal attention maps to project them into the spectral domain. For each attention map $\tilde{\mathcal{A}} \in \{\tilde{\mathcal{A}}^{V \rightarrow T}, \tilde{\mathcal{A}}^{T \rightarrow V}\}$, its centered frequency spectrum $\mathcal{F}_{\text{shift}}(\tilde{\mathcal{A}})$ is computed as:

$$\mathcal{F}_{\text{shift}}(\tilde{\mathcal{A}}) = S(\mathcal{F}(\tilde{\mathcal{A}})), \quad (7)$$

where $\mathcal{F}(\cdot)$ denotes the 2D FFT operator, and $S(\cdot)$ represents the frequency-shifting operation. Subsequently, we decouple the spectrum into distinct frequency bands by applying the Low-Pass Filter ($\mathcal{G}_{\text{LPF}}$) and High-Pass Filter ($\mathcal{G}_{\text{HPF}}$). The spatial low-frequency component $\tilde{\mathcal{A}}_{\text{low}}$ and high-frequency component $\tilde{\mathcal{A}}_{\text{high}}$ are reconstructed via the Inverse 2D FFT ($\mathcal{F}^{-1}$) as follows:

$$\tilde{\mathcal{A}}_{\text{low}} = \text{Re}(\mathcal{F}^{-1}(S^{-1}(\mathcal{F}_{\text{shift}}(\tilde{\mathcal{A}}) \odot \mathcal{G}_{\text{LPF}}))), \quad (8)$$

$$\tilde{\mathcal{A}}_{\text{high}} = \text{Re}(\mathcal{F}^{-1}(S^{-1}(\mathcal{F}_{\text{shift}}(\tilde{\mathcal{A}}) \odot \mathcal{G}_{\text{HPF}}))). \quad (9)$$

where $S^{-1}(\cdot)$ is the inverse shifting operator, $\odot$ denotes the Hadamard product, and $\text{Re}(\cdot)$ extracts the real part of the inverted complex signals. Through this dual-stream spectral decomposition, we ultimately obtain the decoupled low- and high-frequency components, denoted as $\tilde{\mathcal{A}}_{\text{low}}^{V \rightarrow T}$, $\tilde{\mathcal{A}}_{\text{low}}^{T \rightarrow V}$, and $\tilde{\mathcal{A}}_{\text{high}}^{T \rightarrow V}$.

With the decoupled sub-bands, we apply tailored fusion operations to the low- and high-frequency components independently. For the low-frequency bands that primarily capture coarse semantic layouts, we employ a weighted summation to smoothly reconcile the cross-modal global structures:

$$\tilde{\mathcal{A}}_{\text{low}}^{\text{fused}} = w_{\text{low}} \tilde{\mathcal{A}}_{\text{low}}^{V \rightarrow T} + (1 - w_{\text{low}}) \tilde{\mathcal{A}}_{\text{low}}^{T \rightarrow V}, \quad (10)$$

where w_{low} denotes the balancing weight. Meanwhile, for the high-frequency bands that capture fine-grained boundaries, we utilize an element-wise maximum operator to preserve the strongest boundary-aware responses across the two attention fields:

$$\tilde{\mathcal{A}}_{\text{high}}^{\text{fused}} = \max(\tilde{\mathcal{A}}_{\text{high}}^{V \rightarrow T}, \tilde{\mathcal{A}}_{\text{high}}^{T \rightarrow V}). \quad (11)$$

Finally, to obtain the unified spatial representation, we perform spatial recomposition by superimposing the fused high-frequency details onto the low-frequency base. A clipping operation is applied to constrain the fused attention map within a valid range:

$$\tilde{\mathcal{A}}^{\text{fused}} = \text{clamp}(\tilde{\mathcal{A}}_{\text{low}}^{\text{fused}} + \alpha \cdot \tilde{\mathcal{A}}_{\text{high}}^{\text{fused}}, 0, 1), \quad (12)$$

where α represents the high-frequency enhancement scaling factor, and $\text{clamp}(\cdot, 0, 1)$ constrains each element of $\tilde{\mathcal{A}}^{\text{fused}}$ to the range $[0, 1]$.

4.3. Soft mask transition

While the refined attention representation $\tilde{\mathcal{A}}^{\text{fused}}$ provides accurate localization, directly converting it into a spatial mask tends to yield abrupt boundary transitions, resulting in unnatural blending and degrading visual aesthetics. To address this issue, drawing upon local feature learning [43,44], we propose the Soft Mask Transition (SMT) mechanism. SMT propagates semantically consistent regions outward from a localized anchor, yielding a continuous soft mask that seamlessly reconciles the edited regions with the original background.

Specifically, to determine the semantic seed for the k th modified target, we compute a robust intensity-weighted centroid over the most activated spatial positions:

$$S_k = \arg \text{TopK}_{(h,w)}(\tilde{\mathcal{A}}_{h,w,k}^{\text{fused}}), \quad (13)$$

$$\text{centroid}(k) = \frac{\sum_{(h,w) \in S_k} \tilde{\mathcal{A}}_{h,w,k}^{\text{fused}} \begin{bmatrix} h \\ w \end{bmatrix}}{\sum_{(h,w) \in S_k} \tilde{\mathcal{A}}_{h,w,k}^{\text{fused}}}, \quad (14)$$

where $\tilde{\mathcal{A}}_{h,w,k}^{\text{fused}}$ denotes the attention score at spatial coordinate (h, w) for the k th modified target, and $k \in \mathcal{I}$ denotes the index of each modified target. The binary seed mask $\mathbf{M}_k^{(0)} \in \{0, 1\}^{H \times W}$ is initialized by assigning a value of 1 to the coordinate corresponding to $\lfloor \text{centroid}(k) \rfloor$, and 0 elsewhere.

Starting from the initial seed $\mathbf{M}_k^{(0)}$, the mask iteratively propagates outward to its spatial neighbors based on semantic homogeneity. Let $\mathbf{K}_8 \in \{1\}^{3 \times 3}$ denote an all-ones structuring kernel representing the 8-neighborhood, and $*$ be the 2D spatial convolution operator. At each iteration t , we first identify the newly expanded boundary pixels $\mathbf{B}_k^{(t)}$ by subtracting the current mask from its dilated counterpart:

$$\mathbf{B}_k^{(t)} = \mathbb{I}(\mathbf{M}_k^{(t)} * \mathbf{K}_8 > 0) - \mathbf{M}_k^{(t)}, \quad (15)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Then, the mask is updated by incorporating boundary pixels whose attention values satisfy the attention homogeneity criterion:

$$\mathbf{M}_k^{(t+1)} = \mathbf{M}_k^{(t)} + \mathbf{B}_k^{(t)} \odot \mathbb{I}(|\tilde{\mathcal{A}}_k^{\text{fused}} - \mu_k^{(t)}| < \phi), \quad (16)$$

where $\odot$ represents the Hadamard product, ϕ is the propagation threshold, and $\mu_k^{(i)}$ denotes the dynamic mean attention score within the active region of $\mathbf{M}_k^{(i)}$, which is computed as:

$$\mu_k^{(i)} = \frac{\sum_{h,w} \tilde{\mathcal{A}}_{h,w,k}^{\text{fused}} \cdot \mathbf{M}_k^{(i)}(h,w)}{\sum_{h,w} \mathbf{M}_k^{(i)}(h,w) + \epsilon}, \quad (17)$$

where $\epsilon = 10^{-8}$ prevents division by zero. The expansion terminates upon convergence, yielding $\mathbf{M}_{k,\text{final}}$. The $\mathbf{M}_{k,\text{final}}$ is further refined and transformed into a soft transition mask through attention-guided weighting and Gaussian smoothing:

$$\mathbf{M}_k^S = \text{clamp} \left(\mathcal{K} * \left(\hat{\mathbf{M}}_k \odot \sigma \left(10 \cdot (\tilde{\mathcal{A}}_k^{\text{fused}} - 0.5) \right) \right), 0.0, 1.0 \right), \quad (18)$$

where $\sigma(\cdot)$ denotes the sigmoid function, and $\mathcal{K}$ represents a 2D Gaussian kernel. Following the attention scaling configuration adopted in prior methods [14,45,46], we use a scaling factor of 10 and an offset of 0.5 throughout all experiments. The resulting $\mathbf{M}_k^S \in [0, 1]^{H \times W}$ serves as the soft transition mask for the k th modified target. To aggregate the editing regions across all targets, the ultimate unified soft mask $\mathbf{M}^S \in [0, 1]^{H \times W}$ is obtained via an element-wise maximum operation:

$$\mathbf{M}^S = \max_{k \in \mathcal{I}} (\mathbf{M}_k^S). \quad (19)$$

4.4. Integration with training-free pipelines

To achieve seamless aggregation with other training-free methods, we employ a straightforward feature injection:

$$\hat{\mathbf{V}}^{\text{img}} = \mathbf{M}^S \odot \mathbf{V}_{\text{tar}}^{\text{img}} + (\mathbf{1} - \mathbf{M}^S) \odot \mathbf{V}_{\text{src}}^{\text{img}}, \quad (20)$$

where $\hat{\mathbf{V}}^{\text{img}}$ denotes the updated target visual Value matrix, $\mathbf{M}^S$ represents the soft transition mask, and $\mathbf{V}_{\text{src}}^{\text{img}}$ and $\mathbf{V}_{\text{tar}}^{\text{img}}$ denote the intermediate visual Value matrices ($\mathbf{V}_v$ in Section 3.2) within the MM-DiT block of the source and target branches, respectively.

Owing to its highly compatible design, SMART-Edit requires no additional training and refines semantic localization during inference without extra forward passes, making it highly computationally efficient. Moreover, SMART-Edit functions as a plug-and-play module that can be integrated into other training-free editing paradigms:

- **Inversion-based Editing Frameworks:** In these pipelines, $\mathbf{M}^S$ dynamically regulates feature injection between the inversion and denoising branches. It explicitly prevents features associated with the source object from being injected into the denoising process, thereby facilitating the generation of the target content within the edited region while faithfully preserving the unedited background information captured during inversion. The complete workflow is summarized in Algorithm 1 in Appendix A.
- **Inversion-free Editing Frameworks:** These paradigms perform editing through the velocity transition field v^A , constructed from v^{src} and v^{tar} . Guided by $\mathbf{M}^S$, the velocity field within the edited region is progressively shifted from the source trajectory toward the target trajectory, enabling effective semantic replacement. Meanwhile, the velocity field in the unedited regions remains aligned with v^{src} , ensuring the strict retention of the original background. The algorithm pipeline is presented in Algorithm 2 in Appendix A.

5. Experiments

5.1. Experimental setup

Backbones and Datasets. We evaluate SMART-Edit on the image editing task using mainstream flow matching models (SD3-Medium, SD3.5-Medium [39], and Flux.1 Dev [47]). Specifically, to align with the original implementation, SD3-Medium is exclusively utilized for FlowEdit on PIE-Bench, while SD3.5-Medium and Flux.1 Dev are adopted

for the remaining experiments. We evaluate our method on PIE-Bench [27], which utilizes 700 masked images to assess local editing fidelity and background preservation across 9 dimensions, and EditEval [48], which covers 7 common editing tasks with 150 high-quality images.

Evaluation Metrics. Following prior work [27,33,49], we evaluate PIE-Bench across three dimensions: structural integrity (Structure Distance based on DINO [50]), background preservation (PSNR, LPIPS, MSE, and SSIM measured exclusively on unedited regions), and text-to-image alignment (Whole and Edited region CLIP similarity [51]). Furthermore, ImageReward [52] and AES score are incorporated to further assess the visual quality and aesthetic appeal. Similarly, for EditEval, we evaluate editing performance using textual semantic alignment (CLIP-T), content consistency (CLIP-I [51]), and structural integrity (DINO [50]).

Hyperparameter Configurations. In our experiments, we use the official implementations and default hyperparameters for those comparative methods as in [12]. For all three models, we set timesteps to 28. We set the CFG scale to 3.5 for SD3-Medium and SD3.5-Medium, and 2.5 for Flux.1 Dev. SMART-Edit is applied only during the first 4 timesteps for both inversion-based and inversion-free methods. For SD3-Medium and SD3.5-Medium, we apply SMART-Edit to the joint attention maps in the final 12 blocks (indices 12–23). For Flux.1 Dev, we apply SMART-Edit to the attention maps in all 38 single-stream blocks. Notably, SMART-Edit operates in a layer-wise manner, with all submatrices averaged across attention heads before processing. The filter bandwidth and mask boundary sharpness are directly set to $d = 3$ and $k = 5$, respectively. For spectral decomposition, $\mathcal{G}_{\text{LPF}}$ is implemented as a standard 2D Gaussian low-pass filter with a bandwidth parameter $d_s = 0.3$, while its high-pass counterpart is straightforwardly obtained via $\mathcal{G}_{\text{HPF}} = 1 - \mathcal{G}_{\text{LPF}}$. For centroid estimation, the semantic seed is computed over the 10 most activated spatial positions to improve robustness against noisy attention responses. We set $\alpha = 1.2$ for high-frequency component weighting to enhance structural boundaries while preserving consensus. Concurrently, the low-frequency blending weight w_{low} is set to 0.7, while the mask propagation threshold ϕ is configured as 0.10 for Flux.1 Dev and 0.15 for SD3/3.5 to maintain optimal semantic localization precision and prevent semantic leakage. Our experiments are all conducted on a single NVIDIA A100 40 GB GPU. The details of attention extraction are provided in Appendix B. The sensitivity analysis of the five key hyperparameters, including the high-frequency weighting factor α , low-frequency blending weight w_{low} , mask propagation threshold ϕ , injection start step t_s , and injection layer L , is provided in Appendix D.

Baselines. To verify the effectiveness of SMART-Edit, we integrate it with existing methods (FlowEdit [11], RF-Solver [10], and DNAEdit [12]) and compare their image editing capabilities against training-free diffusion-based methods (PnP [32], MasaCtrl [13], DI+PnP [27], DI+MasaCtrl [27], InfEdit [53], InsP2P [54]), and flow-based methods (RF-Inv [9], FTEdit [30], FireFlow [29], FlowEdit [11], RF-Solver [10], CausalCtrl [33], FSI-Edit [8], and FIA-Edit [7]).

5.2. Quantitative results

To evaluate the effectiveness and versatility of SMART-Edit, we conduct comprehensive quantitative experiments across multiple benchmarks. The image editing and quality assessment results on PIE-Bench are reported in Table 1 and the top row of Fig. 4, while the corresponding results on EditEval are summarized in Table 2 and the bottom row of Fig. 4. The results demonstrate that SMART-Edit can be seamlessly integrated into existing training-free, flow-matching image editing frameworks as a plug-and-play module, improving performance across different generative backbones. Equipped with SMART-Edit, all baseline methods achieve a better balance between text-image alignment and structural preservation, while maintaining strong editing capability. In particular, SMART-Edit significantly improves structure preservation, background retention, and semantic editing fidelity, while

Table 1

Quantitative comparison on PIE-Bench. Markers †, ♦, and ‡ indicate results from [12], original papers, and our reproduction from the respective repositories, respectively. Red, blue, and orange cell backgrounds indicate the first, second, and third best performances, respectively. Percentage gains are marked in red.

Method	Model	Structure	Background Preservation				CLIP Similarity		Rank
		Distance [†] ×10 ³	PSNR [†]	LPIPS [†] ×10 ³	MSE [†] ×10 ⁴	SSIM [†] ×10 ²	Whole [†]	Edited [†]	Avg.↓
PnP [†]	SD1.5	27.35	22.31	112.76	82.95	79.25	25.41	22.52	17.86
MasaCtrl [†]	SD1.4	27.12	22.19	105.44	86.37	79.91	24.03	21.15	19.36
DI+PnP [†]	SD1.5	23.35	22.46	105.51	79.94	79.88	25.49	22.64	15.71
DI+MasaCtrl [†]	SD1.4	23.58	22.68	87.41	80.63	81.51	24.39	21.41	17.00
InfEdit [†]	LCM	19.31	27.31	56.32	47.80	85.30	24.90	22.14	10.86
InsP2P [†]	InsP2P	58.13	20.80	159.23	221.30	76.47	23.61	21.68	22.29
RF-Inv [†]	FLUX	42.29	20.20	179.54	139.20	69.91	24.57	22.20	21.71
FTEdit [†]	SD3.5	18.17	26.62	80.55	40.24	91.50	25.74	22.27	8.43
FireFlow [†]	FLUX	29.03	23.33	133.40	70.83	81.22	26.19	22.99	14.57
FSI-Edit [♦]	SD3.5	13.71	26.61	85.44	36.50	86.25	25.69	22.50	9.00
FIA-Edit [♦]	SD3.5	10.34	27.32	55.02	28.66	89.21	25.89	22.82	4.57
FlowEdit [†]	FLUX	27.82	21.96	112.19	94.99	83.08	25.25	22.58	18.00
FlowEdit _{+Ours}	FLUX	24.89 ^{10.5%↓}	23.82 ^{8.5%↑}	80.68 ^{28.1%↓}	71.81 ^{24.4%↓}	85.52 ^{2.9%↑}	25.63 ^{1.5%↑}	22.85 ^{1.2%↑}	12.64
FlowEdit [†]	SD3	27.12	22.22	104.12	85.96	93.22	26.53	23.57	11.21
FlowEdit _{+Ours}	SD3	20.17 ^{25.6%↓}	26.74 ^{20.3%↑}	73.26 ^{29.6%↓}	57.39 ^{33.2%↓}	91.03 ^{2.3%↓}	26.62 ^{0.3%↑}	24.32 ^{3.2%↑}	5.71
RF-Solver [†]	FLUX	21.79	24.83	113.15	52.46	83.38	25.57	22.26	14.57
RF-Solver _{+Ours}	FLUX	17.74 ^{18.6%↓}	26.34 ^{6.1%↑}	74.70 ^{34.0%↓}	45.54 ^{13.2%↓}	87.12 ^{4.5%↑}	26.43 ^{3.4%↑}	23.85 ^{7.1%↑}	6.43
RF-Solver [†]	SD3.5	20.73	21.12	116.39	65.71	83.34	26.15	22.59	14.57
RF-Solver _{+Ours}	SD3.5	18.37 ^{11.4%↓}	25.42 ^{20.4%↑}	68.15 ^{41.5%↓}	48.17 ^{26.7%↓}	85.57 ^{2.7%↑}	27.01 ^{3.3%↑}	23.60 ^{4.5%↑}	6.86
DNAEdit [†]	FLUX	18.87	24.99	95.06	50.45	85.71	25.79	22.87	10.14
DNAEdit _{+Ours}	FLUX	14.94 ^{20.8%↓}	26.79 ^{7.2%↑}	49.83 ^{47.6%↓}	39.79 ^{21.1%↓}	89.65 ^{4.6%↑}	26.33 ^{2.1%↑}	23.35 ^{2.1%↑}	4.43
DNAEdit [†]	SD3.5	14.19	26.66	74.57	32.76	88.63	25.63	22.71	7.36
DNAEdit _{+Ours}	SD3.5	13.42 ^{5.4%↓}	26.68 ^{0.1%↑}	44.56 ^{40.2%↓}	20.73 ^{36.7%↓}	89.49 ^{1.0%↑}	26.88 ^{4.9%↑}	23.83 ^{4.9%↑}	2.71

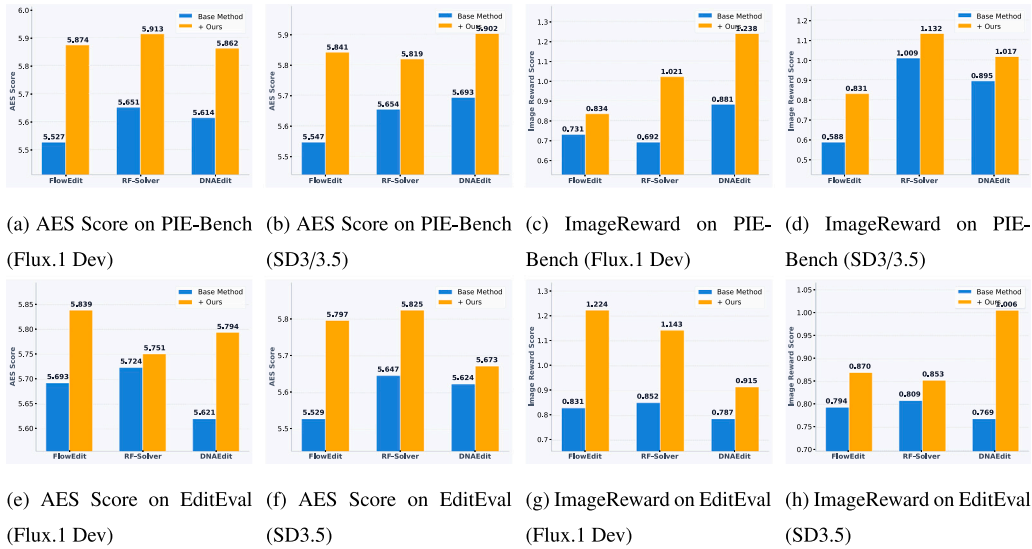


Fig. 4. Image quality assessment results on PIE-Bench (top row) and EditEval (bottom row) benchmarks. From left to right, the columns represent the Aesthetic Score (AES Score) for Flux.1 Dev, AES Score for SD3.5, ImageReward for Flux.1 Dev, and ImageReward for SD3.5, respectively.

also enhancing aesthetic quality. While some individual metrics do not consistently improve, these variations reflect the inherent trade-offs among different evaluation criteria. Conventional editing methods may prioritize strong alignment with the editing prompt, which can lead to modifications beyond the intended editing regions. Such undesired changes can degrade background preservation and introduce inconsistencies between the foreground and background, ultimately reducing the overall visual quality. Conversely, overly conservative background preservation may limit semantic alignment with the editing prompt. By explicitly constraining the editing regions using semantic localization, SMART-Edit achieves a better balance between effective semantic editing and faithful preservation of non-target regions. Moreover, more accurate localization can reduce undesired modifications and editing artifacts around the target regions, which may further contribute to improved visual and aesthetic quality. These results collectively validate

the effectiveness and generalizability of SMART-Edit. We provide the complexity analysis in Appendix E.

5.3. Qualitative results

To intuitively illustrate the advantages of SMART-Edit, we perform a comprehensive qualitative comparison. The qualitative results on PIE-Bench are shown in Fig. 5, while the corresponding visual comparisons on EditEval are presented in Fig. 6. The visual results indicate that existing image editing methods often retain undesired source content when attempting to preserve structural information, while simultaneously introducing distortions into unedited background regions. In contrast, SMART-Edit explicitly separates the edited regions from the surrounding background content, enabling high-fidelity semantic modifications while maintaining strong structural and background consistency. Integrating SMART-Edit into various mainstream frameworks

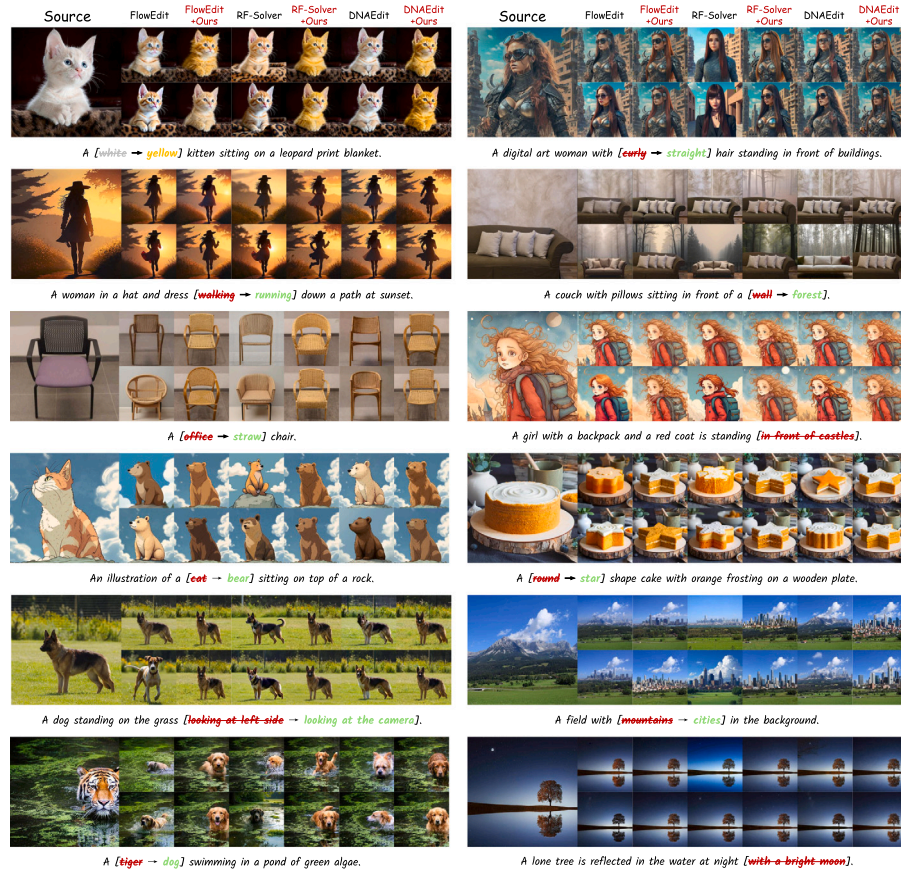


Fig. 5. Qualitative comparison of image editing results on the PIE-Bench dataset. For each case, the top and bottom rows display the results based on SD3/3.5 and Flux.1 Dev, respectively.



Fig. 6. Qualitative comparison of image editing results on the EditEval benchmark. For each case, the top and bottom rows show the results based on SD3/3.5 and Flux.1 Dev, respectively.

Table 2

Quantitative comparison on EditEval. † denotes results copied from [33], and ‡ denotes our reproduced results using the official code. Red, blue, and orange indicate the first, second, and third best performances, respectively.

Method	Model	EditEval		
		CLIP-T †	CLIP-I †	DINO †
RF-Inv†	FLUX	29.3985	0.8554	0.8792
RF-Inv‡	SD3.5	29.5763	0.8672	0.8780
CausalCtrl†	FLUX	30.6173	0.8967	0.9175
FireFlow†	FLUX	29.1872	0.8724	0.8990
FireFlow‡	SD3.5	27.8192	0.8576	0.8737
FlowEdit†	FLUX	30.0495	0.8912	0.9160
FlowEdit _{+Ours}	FLUX	30.1276	0.9267	0.9162
FlowEdit‡	SD3.5	28.3188	0.8608	0.8916
FlowEdit _{+Ours}	SD3.5	30.6755	0.8819	0.8997
RF-Solver†	FLUX	29.2326	0.8884	0.9113
RF-Solver _{+Ours}	FLUX	29.5187	0.9039	0.9189
RF-Solver‡	SD3.5	29.8798	0.8756	0.9048
RF-Solver _{+Ours}	SD3.5	29.7936	0.8974	0.9132
DNAEdit‡	FLUX	29.8497	0.8697	0.8912
DNAEdit _{+Ours}	FLUX	30.6825	0.8992	0.9294
DNAEdit†	SD3.5	30.1159	0.8749	0.9033
DNAEdit _{+Ours}	SD3.5	30.7471	0.9170	0.9276

Table 3

Ablation study of SMART-Edit on PIE-Bench. We report background preservation, editing fidelity, and aesthetic quality metrics on the PIE-Bench dataset.

Model	Method Variants	Distance _{KL} ↓	PSNR↑	Edited↑	AES↑	ImageReward↑
FLUX	DNAEdit _{+Ours}	14.94	26.79	23.35	5.862	1.238
	w/o $\mathcal{A}^{T \rightarrow V}$	16.69	25.27	23.04	5.821	1.155
	w/o $\mathcal{A}^{V \rightarrow T}$	17.20	25.03	22.99	5.772	1.040
	w/o DAR	16.82	26.71	23.10	5.764	1.073
	w/o SMAF	18.56	25.44	23.22	5.798	0.973
	w/o Centroid	15.23	26.34	23.24	5.827	1.156
	w/o SMT	15.03	26.73	23.01	5.562	0.763
	DNAEdit	18.87	24.99	22.87	5.614	0.881
SD3.5	DNAEdit _{+Ours}	13.42	26.68	23.83	5.902	1.017
	w/o $\mathcal{A}^{T \rightarrow V}$	13.48	26.57	23.49	5.779	0.930
	w/o $\mathcal{A}^{V \rightarrow T}$	13.85	26.55	22.97	5.798	0.964
	w/o DAR	13.56	26.59	23.74	5.760	0.975
	w/o SMAF	14.04	26.47	23.08	5.783	0.932
	w/o Centroid	13.67	26.66	23.66	5.817	1.046
	w/o SMT	13.54	26.79	22.98	5.684	0.910
	DNAEdit	14.19	26.66	22.71	5.693	0.895

leads to more accurate semantic edits and improved visual quality, further demonstrating the effectiveness and general applicability of our framework.

5.4. Ablation study

In this section, we present the ablation study results on DNAEdit to evaluate the effectiveness of each component. The ablation results on FlowEdit and RF-Solver are provided in Appendix C.

Effect of Cross-Attention $\mathcal{A}^{V \rightarrow T}$ and $\mathcal{A}^{T \rightarrow V}$. To study the contribution of dual cross-modal attention mechanisms, we remove either $\mathcal{A}^{T \rightarrow V}$ (w/o $\mathcal{A}^{T \rightarrow V}$) or $\mathcal{A}^{V \rightarrow T}$ (w/o $\mathcal{A}^{V \rightarrow T}$) from the mask generation process. As shown in Table 3 and Fig. 7, excluding either attention branch leads to noticeable performance degradation, indicating that the two branches provide complementary localization information. Moreover, the larger performance drop caused by removing $\mathcal{A}^{V \rightarrow T}$ suggests that image-to-text attention plays a more critical role in identifying target editing regions, while text-to-image attention offers complementary semantic guidance.

Effect of DAR. We evaluate the effectiveness of DAR by removing the self-attention-based refinement and directly using the original cross-modal attention maps for semantic localization (w/o DAR). As shown in Table 3 and Fig. 7, removing DAR degrades performance across

both Flux.1 Dev and SD3.5. This result demonstrates the importance of self-attention refinement for generating accurate editing masks. By suppressing spurious activations and enhancing spatial coherence, DAR provides more reliable localization cues for text-guided image editing.

Effect of SMAF. We evaluate SMAF by directly fusing $\mathcal{A}^{V \rightarrow T}$ and $\mathcal{A}^{T \rightarrow V}$ through weighted summation, without frequency decomposition and sub-band fusion. As shown in Table 3 and Fig. 7, performance drops after removing SMAF, demonstrating that the inherent spectral discrepancies between the two cross-modal attentions cannot be adequately reconciled by a naive fusion approach. In contrast, SMAF explicitly aligns semantic structures and boundary information across different frequency bands, resulting in more reliable editing localization and superior editing quality.

Effect of Centroid and SMT. We further investigate the proposed centroid initialization and Soft Mask Transition (SMT). Specifically, “w/o Centroid” replaces the attention-weighted centroid with a fixed initialization at the global maximum response of the attention map as the semantic seed, while “w/o SMT” directly converts the fused attention map into a mask without the proposed region propagation and soft transition mechanisms. As shown in Table 3 and Fig. 7, removing either component degrades performance across both backbones. The results indicate that the centroid provides a reliable semantic anchor for region expansion, while SMT further improves the spatial continuity and boundary smoothness of the generated masks. Such improvements are particularly important for producing visually coherent edits and reducing boundary artifacts, thereby contributing to higher perceptual image quality. Together, these two components facilitate more accurate region localization and more natural editing transitions, leading to improved visual fidelity.

5.5. Visualization analysis

In this section, we present cross-attention visualizations for both the inversion and denoising processes to provide qualitative evidence of how SMART-Edit achieves precise text-to-image semantic alignment. All attention maps are computed by averaging attention weights across all transformer layers and heads for visualization stability. Specifically, we visualize cross-attention maps at the final timestep of the denoising process. For the inversion process, we present the visualizations at the initial timestep.

Precise Semantic Localization During Inversion. As shown in Fig. 8, traditional methods that directly threshold image-to-text cross-attention often suffer from semantic leakage, leading to imprecise localization and unintended edits. Our visualizations demonstrate how SMART-Edit addresses this limitation through a progressive refinement pipeline. First, the DAR module refines the dual cross-attention maps to suppress abnormal activations. The SMAF module then fuses the refined attentions in the frequency domain to produce a more accurate and spatially coherent semantic localization map, which is subsequently transformed into editing-friendly masks by the SMT module. Together, these components effectively confine attention activations to the intended editing regions, enabling accurate semantic modifications while preserving the surrounding background content.

Multi-Keyword Precise Localization During Inversion. As illustrated in Fig. 9, SMART-Edit is capable of accurately localizing multiple semantic keywords simultaneously within a single timestep, while conventional inversion methods often struggle to disentangle multiple entities, resulting in ambiguous localization or overlapping attention responses. This limitation becomes particularly problematic in complex editing scenarios that involve multiple objects and their spatial relationships. The visualizations show that SMART-Edit effectively separates the attention responses associated with different semantic concepts, enabling each keyword to be consistently anchored to its corresponding spatial region. Consequently, the generated masks exhibit clear spatial boundaries with minimal cross-interference between semantic entities.

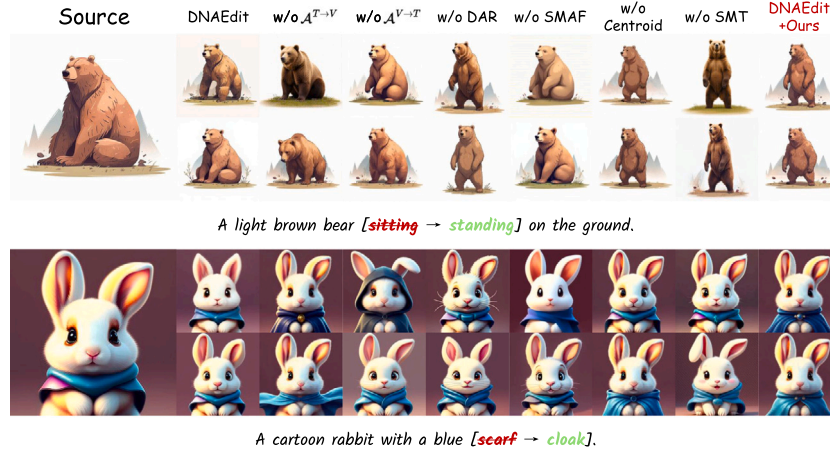


Fig. 7. Qualitative comparison of ablation variants. For each case, results generated by SD3.5 and Flux.1 Dev are shown in the first and second rows, respectively. Please zoom in for a better view.

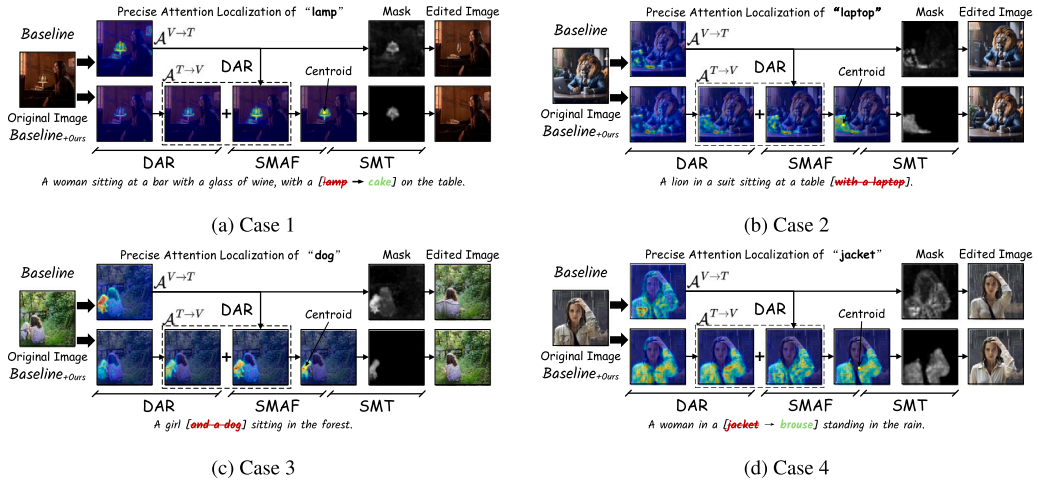


Fig. 8. Visualization of the semantic localization process during inversion. We compare our SMART-Edit with the baseline DNAEdit [12] across four cases. The results show that SMART-Edit progressively refines attention maps into editing-friendly masks, achieving more precise semantic localization.

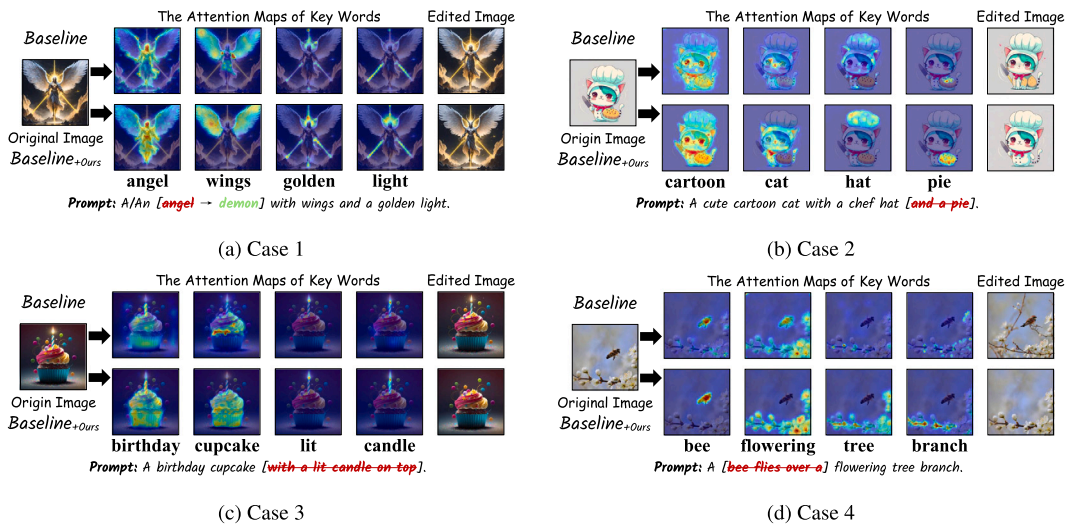


Fig. 9. Multi-keyword attention visualization during inversion. We compare our SMART-Edit with the baseline DNAEdit [12] across four cases. SMART-Edit precisely localizes multiple objects simultaneously, while the baseline produces ambiguous or misaligned attention responses.

This accurate multi-keyword localization facilitates precise semantic editing while helping preserve the integrity of non-target regions.

Attention Visualization During Denoising. As illustrated in Fig. 10, the precise region-specific semantic control established during the

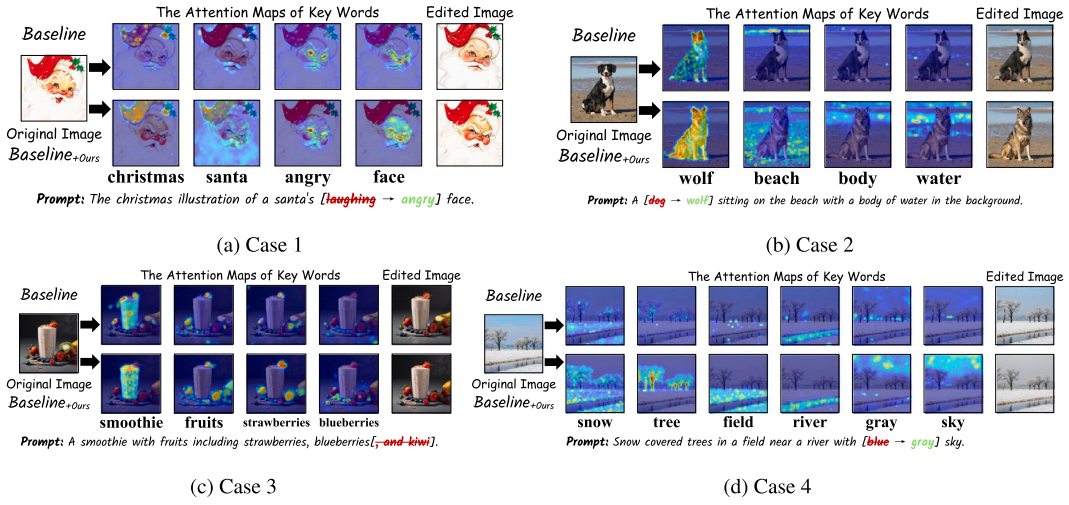


Fig. 10. Attention visualization during denoising. We compare our SMART-Edit with the baseline DNAEdit [12] across four cases. SMART-Edit achieves precise image-to-text semantic alignment via mask optimization, whereas the baseline produces degraded attention responses and weaker text-image alignment.

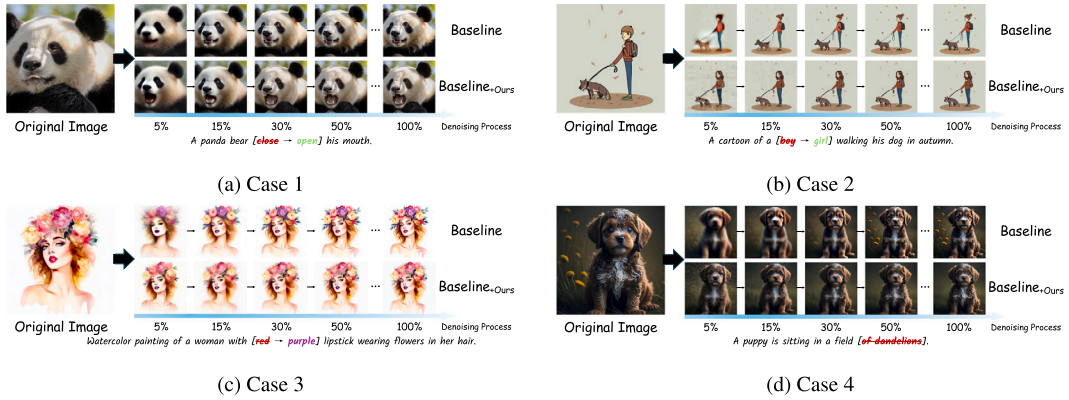


Fig. 11. Latent evolution during denoising. We compare our SMART-Edit with the baseline DNAEdit [12]. SMART-Edit incorporates target semantics at early diffusion steps and stably preserves source semantics throughout denoising, whereas the baseline exhibits semantic drift and fails to maintain consistent source structure. Please zoom in for a better view.

inversion process significantly promotes more accurate image-to-text semantic alignment during the subsequent denoising phase. Inherently, there is a natural semantic binding between the original image and the source prompt. By effectively preserving the visual information of the unedited regions, SMART-Edit successfully sustains this inherent semantic coherence throughout the editing trajectory. Furthermore, during the generative denoising steps, these masks effectively prevent the newly injected semantics of the targeted editing regions from over-diffusing or bleeding into irrelevant background areas. Consequently, this synergistic mechanism guarantees that the model accurately grounds the editing concepts to the designated regions, achieving high-fidelity semantic modifications while perfectly maintaining the contextual integrity of the original image.

Latent Trajectory During Denoising. To gain deeper insight into how SMART-Edit guides latent dynamics during the denoising process, we perform one-step projection at different diffusion timesteps and decode them for visualization, as shown in Fig. 11. We find that non-edited regions converge to visually faithful representations consistent with the source image, even at very early stages, whereas edited regions undergo continuous refinement until the final timestep. This finding corroborates the necessity of our mask-based editing strategy. In contrast, baselines either fail to integrate target semantics at early diffusion steps or inadvertently alter the background structure during semantic injection. SMART-Edit successfully avoids both issues. Notably, we find that the early diffusion stages are particularly critical for achieving

both precise editing and background preservation, and our feature injection strategy strikes a favorable trade-off between editing fidelity and background consistency.

6. Limitations and future work

Similar to existing attention-driven editing methods [13,16], while SMART-Edit excels at semantic deletion and replacement, its performance may be bounded by the underlying baseline methods during semantic addition tasks. Inherently, our framework relies on the premise that the target semantics already exist within the source image. When adding entirely new semantic concepts, the source image lacks corresponding visual features, making it difficult to extract meaningful cross-attention maps for semantic localization. In future work, we plan to alleviate this reliance on explicit token indices, thereby generalizing our framework to a broader range of image editing applications. More detailed failure case analysis and discussions of future directions are provided in Appendix F.

7. Conclusion

In this work, we introduced SMART-Edit, a novel plug-and-play framework for training-free image editing that can be seamlessly integrated into both inversion-based and inversion-free editing paradigms.

By leveraging complementary text-to-image and image-to-text cross-attention signals within MM-DiT, SMART-Edit enables accurate semantic localization and spatially precise editing control through dual-attention refinement and spectral fusion, without requiring any additional training. Extensive experiments across multiple benchmarks and generative backbones demonstrate that SMART-Edit improves editing fidelity, structural preservation, and visual quality while introducing only minimal computational overhead. Beyond its practical effectiveness, SMART-Edit provides a general semantic localization framework tailored to MM-DiT architectures, offering a direction for controllable and training-free image editing.

CRedit authorship contribution statement

Jili Chen: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Yaxin Tu:** Writing – review & editing, Validation, Software, Methodology, Conceptualization. **Qionghao Huang:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Xizhe Wang:** Writing – review & editing, Resources, Investigation, Data curation. **Xiaodi Huang:** Writing – review & editing, Visualization, Validation, Formal analysis. **Changqin Huang:** Writing – review & editing, Writing – original draft, Conceptualization, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2026.115025>.

Data availability

The datasets used in this study are publicly available benchmark datasets. The official implementation of SMART-Edit is publicly available [here](#).

References

- [1] Y. Lipman, R.T.Q. Chen, H. Ben-Hamu, M. Nickel, M. Le, Flow matching for generative modeling, in: International Conference on Learning Representations, 2023.
- [2] W. Peebles, S. Xie, Scalable diffusion models with transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 4195–4205.
- [3] T. Brooks, A. Holynski, A.A. Efros, Instructpix2pix: Learning to follow image editing instructions, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 18392–18402.
- [4] G. Kim, T. Kwon, J.C. Ye, Diffusionclip: Text-guided diffusion models for robust image manipulation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 2426–2435.
- [5] T. Wang, T. Zhang, B. Zhang, H. Ouyang, D. Chen, Q. Chen, F. Wen, Pretraining is all you need for image-to-image translation, 2022, arXiv preprint arXiv:2205.12952.
- [6] T. Zhu, S. Zhang, J. Shao, Y. Tang, Kv-edit: Training-free image editing for precise background preservation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 16607–16617.
- [7] K. Yang, B. Shen, X. Li, Y. Dai, Y. Luo, Y. Ma, W. Fang, Q. Li, Z. Wang, FIA-edit: Frequency-interactive attention for efficient and high-fidelity inversion-free text-guided image editing, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, (14) 2026, pp. 11613–11621.
- [8] K. Yang, X. Li, Y. Li, Q. Li, Z. Wang, FSI-edit: Frequency and stochasticity injection for flexible diffusion-based image editing, in: The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
- [9] L. Rout, Y. Chen, N. Ruiz, C. Caramanis, S. Shakkottai, W.-S. Chu, Semantic image inversion and editing using rectified stochastic differential equations, in: International Conference on Learning Representations, vol. 2025, 2025, pp. 77330–77365.
- [10] J. Wang, J. Pu, Z. Qi, J. Guo, Y. Ma, N. Huang, Y. Chen, X. Li, Y. Shan, Taming rectified flow for inversion and editing, in: Proceedings of the 42nd International Conference on Machine Learning, in: Proceedings of Machine Learning Research, vol. 267, PMLR, 2025, pp. 64044–64058.
- [11] V. Kulikov, M. Kleiner, I. Huberman-Spiegelglas, T. Michaeli, Flowedit: Inversion-free text-based editing using pre-trained flow models, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 19721–19730.
- [12] C. Xie, M. Li, S. Li, Y. Wu, Q. Yi, L. Zhang, Dnaedit: Direct noise alignment for text-guided rectified flow editing, Adv. Neural Inf. Process. Syst. 38 (2026) 124456–124474.
- [13] M. Cao, X. Wang, Z. Qi, Y. Shan, X. Qie, Y. Zheng, Masactrl: Tuning-free mutual self-attention control for consistent image synthesis and editing, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 22560–22570.
- [14] Z. Zhou, Y. Deng, X. He, W. Dong, F. Tang, Multi-turn consistent image editing, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 15792–15801.
- [15] Y. Hu, J. Peng, Y. Lin, T. Liu, X. Qu, L. Liu, Y. Zhao, Y. Wei, Dcredit: Dual-level controlled image editing via precisely localized semantics, 2025, arXiv preprint arXiv:2503.16795.
- [16] Y. Xu, F. Tang, J. Cao, X. Kong, Y. Zhang, J. Li, O. Deussen, T.-Y. Lee, Headrouter: A training-free image editing framework for mm-dits by adaptively routing attention heads, ACM Trans. Graph. 45 (3) (2026) 1–14.
- [17] H. Chefer, Y. Alaluf, Y. Vinker, L. Wolf, D. Cohen-Or, Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models, ACM Trans. Graph. 42 (4) (2023) 1–10.
- [18] P. Dhariwal, A. Nichol, Diffusion models beat gans on image synthesis, Adv. Neural Inf. Process. Syst. 34 (2021) 8780–8794.
- [19] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.
- [20] Y. Ma, X. Wang, Q. Ma, Q. Wang, M. Zheng, X. Yang, H. Li, C. Zhao, J. Ying, H. Yang, et al., Group editing: Edit multiple images in one go, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 43418–43428.
- [21] Z. Zhang, Q. Zhang, W. Xing, G. Li, L. Zhao, J. Sun, Z. Lan, J. Luan, Y. Huang, H. Lin, Artbank: Artistic style transfer with pre-trained diffusion model and implicit style prompt bank, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, (7) 2024, pp. 7396–7404.
- [22] Z. Zhang, Q. Zhang, H. Lin, W. Xing, J. Mo, S. Huang, J. Xie, G. Li, J. Luan, L. Zhao, D. Zhang, L. Chen, Towards highly realistic artistic style transfer via stable diffusion with step-aware and layer-aware prompt, in: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, 2024, pp. 7814–7822.
- [23] Z. Zhang, Q. Zhang, G. Li, J. Luan, M. Yang, Y. Wang, L. Zhao, Dyartbank: Diverse artistic style transfer via pre-trained stable diffusion and dynamic style prompt artbank, Knowl.-Based Syst. 310 (2025) 112959.
- [24] Z. Zhang, Y. Shao, A. Ma, Q. Zhang, H. Tang, N. Sebe, SAST: Artistic style transfer via pre-trained stable diffusion with step-aware content and style learning, IEEE Trans. Multimed. (2026) 1–13.
- [25] J. Song, C. Meng, S. Ermon, Denoising diffusion implicit models, in: International Conference on Learning Representations, 2021.
- [26] I. Huberman-Spiegelglas, V. Kulikov, T. Michaeli, An edit friendly ddpm noise space: Inversion and manipulations, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 12469–12478.
- [27] X. Ju, A. Zeng, Y. Bian, S. Liu, Q. Xu, Pnp inversion: Boosting diffusion-based editing with 3 lines of code, in: International Conference on Learning Representations, vol. 2024, 2024, pp. 23395–23422.
- [28] X. Liu, C. Gong, Q. Liu, Flow straight and fast: Learning to generate and transfer data with rectified flow, in: International Conference on Learning Representations, 2023.
- [29] Y. Deng, X. He, C. Mei, P. Wang, F. Tang, FireFlow: Fast inversion of rectified flow for image semantic editing, in: Proceedings of the 42nd International Conference on Machine Learning, in: Proceedings of Machine Learning Research, vol. 267, PMLR, 2025, pp. 13110–13128.

- [30] P. Xu, B. Jiang, X. Hu, D. Luo, Q. He, J. Zhang, C. Wang, Y. Wu, C. Ling, B. Wang, Unveil inversion and invariance in flow transformer for versatile image editing, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 28479–28489.
- [31] X. Zhang, X. Niu, R. Chen, D. Song, J. Zeng, P. Du, H. Cao, K. Wu, A.-a. Liu, Group relative attention guidance for image editing, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 3840–3850.
- [32] N. Tumanyan, M. Geyer, S. Bagon, T. Dekel, Plug-and-play diffusion features for text-driven image-to-image translation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 1921–1930.
- [33] H. Cao, C. Wang, Y. Lai, S. Min, X. Chen, CausalCtrl: Causality-aware control framework for text-guided visual editing, in: Proceedings of the 33rd ACM International Conference on Multimedia, 2025, pp. 9920–9929.
- [34] J. Wang, X. Li, J. Zhang, Q. Xu, Q. Zhou, Q. Yu, L. Sheng, D. Xu, Diffusion model is secretly a training-free open vocabulary semantic segmenter, IEEE Trans. Image Process. (2025).
- [35] Z. Yin, L.-H. Chen, L. Ni, X. Dai, ConsistEdit: Highly consistent and precise training-free visual editing, in: Proceedings of the SIGGRAPH Asia 2025 Conference Papers, 2025, pp. 1–11.
- [36] J. Li, H. Xu, G. Chen, S. Zhang, Quantum multi-view feature selection with configurable kernel circuits and adaptive fusion, IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst. (2025).
- [37] H. Xu, J. Li, Intertwine: Nonlinear quantum feature selection with multi-kernel circuits, IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst. (2025).
- [38] J. Li, H. Xu, S. Zhang, Adaptive kNN graph model, Nat. Commun. (ISSN: 2041-1723) 17 (1) (2026) 7491.
- [39] P. Esser, S. Kulal, A. Blattmann, R. Entezari, J. Müller, H. Saini, Y. Levi, D. Lorenz, A. Sauer, F. Boesel, et al., Scaling rectified flow transformers for high-resolution image synthesis, in: Forty-First International Conference on Machine Learning, 2024.
- [40] J. Chen, H. Zeng, C. Huang, Q. Huang, Z. Li, X. Huang, Context-aware latent space mediation for inference-time unbiased semantic alignment in text-to-image models, Expert Syst. Appl. (2026) 133019.
- [41] G. Koo, S. Yoon, J.W. Hong, C.D. Yoo, Flexiedit: Frequency-aware latent refinement for enhanced non-rigid editing, in: European Conference on Computer Vision, Springer, 2024, pp. 363–379.
- [42] X. Guo, X. Fu, M. Zhou, Z. Huang, J. Peng, Z.-J. Zha, Exploring Fourier prior for single image rain removal, in: IJCAI, 2022, pp. 935–941.
- [43] J. Xiao, X. Fu, A. Liu, F. Wu, Z.-J. Zha, Image de-raining transformer, IEEE Trans. Pattern Anal. Mach. Intell. 45 (11) (2022) 12978–12995.
- [44] J. Xiao, X. Fu, Y. Zhu, Z.-J. Zha, Bayesian window transformer for image restoration, IEEE Trans. Pattern Anal. Mach. Intell. 48 (3) (2026) 2431–2444.
- [45] D. Epstein, A. Jabri, B. Poole, A. Efros, A. Holynski, Diffusion self-guidance for controllable image generation, Adv. Neural Inf. Process. Syst. 36 (2023) 16222–16239.
- [46] Y. Dalva, K. Venkatesh, P. Yanardag, FluxSpace: Disentangled semantic editing in rectified flow models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 13083–13092.
- [47] B.F. Labs, S. Batifol, A. Blattmann, F. Boesel, S. Consul, C. Diagne, T. Dockhorn, J. English, Z. English, P. Esser, et al., Flux. 1 kontext: Flow matching for in-context image generation and editing in latent space, 2025, arXiv preprint arXiv:2506.15742.
- [48] Y. Huang, J. Huang, Y. Liu, M. Yan, J. Lv, J. Liu, W. Xiong, H. Zhang, L. Cao, S. Chen, Diffusion model-based image editing: A survey, IEEE Trans. Pattern Anal. Mach. Intell. (2025).
- [49] M. Li, C. Xie, Y. Wu, L. Zhang, M. Wang, Five-bench: A fine-grained video editing benchmark for evaluating emerging diffusion and rectified flow models, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 16672–16681.
- [50] M. Oquab, T. Darcet, T. Moutakanni, H.V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jegou, J. Mairal, P. Labatut, A. Joulin, P. Bojanowski, DINOv2: Learning Robust Visual Features Without Supervision, Trans. Mach. Learn. Res. (2024).
- [51] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PmlR, 2021, pp. 8748–8763.
- [52] J. Xu, X. Liu, Y. Wu, Y. Tong, Q. Li, M. Ding, J. Tang, Y. Dong, Imagereward: Learning and evaluating human preferences for text-to-image generation, Adv. Neural Inf. Process. Syst. 36 (2023) 15903–15935.
- [53] S. Xu, Y. Huang, J. Pan, Z. Ma, J. Chai, Inversion-free image editing with language-guided diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 9452–9461.
- [54] A. Haque, M. Tancik, A.A. Efros, A. Holynski, A. Kanazawa, Instruct-nerf2nerf: Editing 3d scenes with instructions, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 19740–19750.